Logistic regression

Use the same example dataset for Kaggle forest health.

Set up

Our set up is the following:
- one quantitative predictor X
- one quantitative predictor Y with two classes: class 1, class 2.

Instead of our classifier directly assigning a class, we want to model $p(Y=1 \mid X) = p(X)$ = probability of Y being class 1, given we observe X.

Then we can make predictions by selecting a threshold α and decide that

if $p(X) \geq \alpha$, then we predict class 1

if $p(X) < \alpha$, then we predict class 0.

so we want to predict whether a Kaiap forest status (healthy / degraded) based on the recorded temperature anomaly.

models the probability of Kaiap forest being healthy given temperature anomaly:

$$P_r(\text{status} = \text{healthy} \mid \text{temperature}) = p(\text{temp})$$

This is "the probability that the status is healthy given a temperature value". The probability lies between 0 and 1.

If we know $p(\text{temp})$ for any value of temp, then we can select a threshold value to do the classification, for example

- if $p(\text{temp}) \geq 0.8 \Rightarrow$ predict site with given temp is healthy
- if $p(\text{temp}) < 0.8 \Rightarrow$ predict site with given temp is degraded

Logistic regression

Instead of modelling which class each observation belongs to directly, logistic regression models the probability that an observation belongs to the positive class.

This model for the probability $p(X)$ is given by the logistic function

$$p(X) = \frac{e^{\beta_0 + \beta_1 X}}{1 + e^{\beta_0 + \beta_1 X}}.$$

Here β_0, β_1 are the coefficients we need to estimate to fit the logistic model.

* How to make classifications with the threshold
* Example with the lessons data.

Maximum likelihood

A method called maximum likelihood is used to fit the logistic model and find the coefficients.

To fit the model we use our training data $\{(x_1, y_1), \dots, (x_n, y_n)\}$

remember $y_i = 0$ or 1.

Our model doing a good job means that if x_i is associated with $y_i = 1$, then the model gives a probability to x_i close to 1,
and if x_i is associated with $y_i = 0$, then the models gives a probability close to 0.

This is equivalent to maximizing the likelihood function

$$\ell(\beta_0, \beta_1) = \prod_{i: y_i = 1} p(x_i) \prod_{i': y_{i'} = 0} (1 - p(x_{i'})).$$

Interpreting the β_1 coefficient

If $\beta_1 > 0$, then increasing X will be associated with increasing $p(X)$.

This means, the larger X, the larger the probability of belonging to class 1.

If $\beta_1 < 0$, then increasing Y...

This means, the larger Y, the smaller the probability of belonging to class 0.

Why?

The logistic regression models the probability of belonging to class 1 as

$$p(X) = \frac{e^{\beta_0 + \beta_1 X}}{1 + e^{\beta_0 + \beta_1 X}}.$$

This can be re-arranged into the odds

$$\frac{p(X)}{1 - p(X)} = e^{\beta_0 + \beta_1 X}.$$

The odds can take any value between 0 and ∞.
* Include how to interpret odds with numbers

And taking logarithm on both sides we obtain the log-odds

$$\log\left(\frac{p(X)}{1 - p(X)}\right) = \beta_0 + \beta_1 X.$$

This tells us that the log odds are linear in X.

Also:
If the log odds increase, so does $p(X)$, and in this formula we can see log odds will increase if $\beta_1 > 0$.

Similarly, the log odds will decrease if $\beta_1 < 0$, and if the log odds decrease, so does $p(X)$.

Hypothesis testing

Once we have estimated the regression coefficients, we can ask whether there is really a relationship between X and Y.

Remember $p(X) = \dfrac{e^{\beta_0 + \beta_1 X}}{1 + e^{\beta_0 + \beta_1 X}}$,

so if $X = 0$, $p(X)$ will be constant, meaning the probability of the positive class does not depend on X.

The null hypothesis is then

$H_0: \beta_1 = 0 \Rightarrow$ the prob of the positive class does not depend on X

$H_a: \beta_1 \neq 0 \Rightarrow \dots$

We test H_0 using the z-statistic associated with β_1 and computing a p-value. If the p-value is small enough it provides evidence against the null hypothesis.

Check in with lessons data about interpreting coeffs and hypothesis testing.

Accuracy assessment

Link with previous lesson on error rate and confusion matrix. Probably have an example with the logistic regression model applied to the Kaiap data with a 50% probability threshold

Adjusting the threshold

Remember the logistic regression models the probability of an observation belonging to the positive class.
* want to change class 1/0 for positive/negative class to make it less confusing

and the observation gets assigned the positive class if the probability of belonging to the positive class exceeds an train threshold:

if $p(X) \geq \alpha$, then X gets assigned positive class

if $p(X) < \alpha$, then X gets assigned negative class

Check in example interpreting whether the classification is more or less critical in the examples data with $\alpha = 0.9$ and $\alpha = 0.2$, and what types of error (false positives or false negatives) we may expect in each case.

Show equivalent of Figure ISLR 4.7 with the error rate and the sensitivity / specificity as a function of the threshold.

* How can we select a best threshold?
Decision must be associated with domain knowledge and an understanding of the trade offs between errors and their implications.

ROC curve

ROC curve stands for receiver operator characteristics, coming from?
Shows the performance of a binary classifier model across threshold values.
It is drawn by calculating the true positive rate (TPR) and false positive rate (FPR) at every possible threshold, then graphing TPR over respect to FPR.
If we want to draw it by hand it would be something like:

threshold	means	false positive rate $\frac{FP}{FP+TN} = \frac{FP}{N}$	true positive rate $\frac{TP}{TP+FN} = \frac{TP}{P}$
0	we classify everything as 1	1	1
1	we classify everything as N	0	0

Include example of something in between. Maybe do this on a desk in with two new variables of threshold and having these be on a table, we populate a simple one by hand.

"* ROC curves show that on steep in the beginning corresponds to thresholds where the tradeoff between FPR and TPR is worth it.
ROC curves that look like a diagonal from (0,0) to (1,1) mean that you have to pick up a lot of true positives to reduce your false positives."
* Need to add false positive rate to KNN accuracy notes.

Show the plot from wikipedia with the different ROC interpretations.

Area under the curve (AUC)

A spam classifier with AUC of 1 means:
"Assigns a random spam email a higher probability of being spam than a random legitimate email. (?) Actual classification depends on the selected threshold."